

Trader Behavior vs Market Sentiment Analysis

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1. Executive Summary

Cryptocurrency markets are highly driven by human emotions, particularly fear and greed. This project analyzes how trader behavior and performance change across different market sentiment regimes using historical trade-level data combined with the Bitcoin Fear & Greed Index. By linking individual trades with daily market sentiment, the analysis explores how profitability, risk-taking, trading activity, and leverage usage vary under different emotional market conditions.

2. Objective

The objective of this project is to examine the relationship between market sentiment (Fear vs Greed) and trader behavior. Specifically, the study aims to understand how trader profitability, risk exposure, trading volume, and leverage usage align or diverge from overall market sentiment, and to identify behavioral patterns that can support smarter trading decisions.

3. Datasets Used

Bitcoin Market Sentiment Dataset

This dataset provides daily classifications of Bitcoin market sentiment such as Extreme Fear, Fear, Neutral, Greed, and Extreme Greed. It represents the collective emotional state of market participants rather than individual actions.

Historical Trader Data (Hyperliquid)

This dataset contains detailed trade-level records, where each row represents a single executed trade. It includes information such as trade size, execution price, trade direction, starting position, transaction fees, and realized profit or loss. This data reflects real trading behavior in live market conditions.

4. Methodology

The analysis began with data cleaning and preprocessing, including standardizing column names, converting timestamps to datetime format, and removing invalid trades. Trade

execution dates were extracted and used to merge trader data with daily market sentiment using a left join, ensuring no loss of trading information.

Since explicit leverage information was not consistently available, a proxy leverage metric was created using trade size relative to existing positions. Exploratory Data Analysis (EDA) was then performed to study how profitability, risk, volume, and leverage vary across different sentiment regimes.

5. Interpretation

Trader profitability varies across market sentiment regimes, with emotionally extreme phases showing higher variability in outcomes.

Fear-driven markets often encourage larger but more disciplined trade sizes, suggesting opportunistic or contrarian behavior.

Trading activity increases during fear and greed phases, while extreme fear is associated with reduced market participation.

Leverage usage tends to increase during greed-driven markets, reflecting higher risk appetite.

Excessive leverage does not consistently improve profitability and often amplifies losses.

6. Conclusion

The findings suggest that trading strategies should adapt to market sentiment. Reducing leverage during extreme greed phases and maintaining disciplined risk exposure during fearful markets can improve risk-adjusted performance. Incorporating sentiment indicators into trading frameworks can help traders avoid emotionally driven decisions and manage risk more effectively.

This project demonstrates that market sentiment plays a significant role in shaping trader behavior and performance. By integrating trader-level data with sentiment indicators, the analysis reveals emotional inefficiencies that impact risk-taking and profitability.